Commercial processes

Higher · Materials & manufacture · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ describe the uses and identifying features of injection moulding, die casting, rotational moulding, vacuum forming and sand casting
- ☐ describe the features of a good vacuum-forming and sand-casting pattern
- ☐ describe the uses and identifying features of bending, blow moulding, compression moulding, drop forging, extrusion, piercing and blanking, and press forming
- ☐ describe the issues that influence the choice of process
- ☐ describe how to identify the process used to make a product

I DO — Worked example

Read how a full-mark answer is built

A large plastic storage box with a hollow, double-skinned lid.

1 Name the process used to make the hollow lid and outline two reasons it is suitable. (3 marks)

Full-mark answer each ✓ is one mark

Rotational moulding ✓. It produces a hollow, seamless lid in one piece ✓ with an even wall thickness and a good surface finish ✓.

Why marks are not awarded

“Injection moulding because it is good for mass production.”

0 marks: injection moulding cannot make a closed hollow form, and "mass production" repeats the question.

WE DO — Guided practice

Use the starters and word bank to help you

A cast iron garden weight.

2 Describe two features that show the weight was sand cast. (2 marks)

Sentence starters

First, …

Next, … using a …

Word bank

rounded · texture · fettling · runner · surface · fillets · angles · corners

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Plastic window frames with an identical profile along their length are made by...
☐ Blow moulding ☐ Extrusion ☐ Compression moulding ☐ Drop forging

b) A spanner needs to be very strong. Process?
☐ Sand casting ☐ Drop forging ☐ Laser cutting ☐ Vacuum forming

c) Small circular marks on the inside of a plastic casing are evidence of...
☐ Ejector pins ☐ Sand grains ☐ Flashing ☐ Layer lines

d) A plastic kayak is hollow, seamless and has an even wall thickness. Process?
☐ Injection moulding ☐ Vacuum forming ☐ Rotational moulding ☐ Die casting

4 Key terms: write the letter of the matching description next to each term.

1. **Injection moulding**	____	**A** Large hollow seamless plastic forms: kayaks, bins.
2. **Die casting**	____	**B** Thermoplastic sheet pulled over a pattern: thin walls, draft angles.
3. **Rotational moulding**	____	**C** Allow trapped air out of hollows in a vacuum-forming pattern.
4. **Sand casting**	____	**D** Low-volume metal parts with rough sandy texture.
5. **Vacuum forming**	____	**E** High-volume complex metal parts in aluminium/zinc. Same features as injection moulding.
6. **Vent holes**	____	**F** High-volume complex thermoplastic parts. Ejector pin marks, injection point, flash.

A games controller casing.

5 State two features that show the casing was injection moulded. (2 marks)

A chocolate box insert.

6 State two features that show the insert was vacuum formed. (2 marks)

Zinc alloy toy cars.

7 Outline two reasons die casting is suitable for the toy cars. (2 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?